

Scheduling Logic: Department vs Description, Rating and Improvement Plan

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Companion to the PPM and CalSchedule deep dive. Code read in full; live data not yet probed because the dev database was down (section 6).

Verdict

Description logic scores 3.90 / 5 (78%); department logic scores 3.05 / 5 (61%). Make description the default for both PPM and calibration, and let each workshop switch to department when ward access is its real constraint.

The deciding reason is the **interval**. An interval belongs to a *type* of device, not to a ward. A description group always shares one cycle; a department group rarely does.

The current system scores 3.5 / 15 (23%) against the fifteen business rules. The improvement plan in section 5 takes it to 15 / 15 in five phases. Phase 0 is two days of low-risk fixes that stop history being lost and equipment going unscheduled.

1. What was decided

Scheduling is configured **per workshop**. Each workshop picks **one** of two logics, separately for PPM and for calibration:

Logic	A group is	The administrator sets	Example
Department	All devices in one ward	Months per department	ICU: January, July
Description	All devices of one type	Months per description	Patient Monitor: February, August

One logic per workshop means a device matches exactly one rule. The "conflicting rules" case disappears, and every schedule can be explained in a single sentence.

2. Department vs description, scored

Each criterion is scored 1 to 5 (5 is best) and weighted by how much it matters for a biomedical workshop running 5,000+ devices. Weights are a judgement; the table shows them so they can be argued with.

Criterion	Weight	Dept	Desc	Department	Description
Interval consistency inside a group	20%	2	5	A ward mixes 3-, 6- and 12-month devices, so one department cannot share one cycle.	Every device of one type shares an interval, so a group has one cycle.
Batching the actual work	15%	2	5	Each visit needs every test instrument, procedure and spare kit.	One procedure, one analyser and one skill set per batch. Strongest for calibration.
Ward access and clinical disruption	15%	5	2	One visit per ward per cycle, and one handover with the nurse in charge.	Engineers return to the same ward several times a year.
Monthly workload balance	15%	2	4	Few, uneven groups. ICU alone can be 300+ devices landing in one month.	Many smaller groups can be spread evenly across the allowed months.

Criterion	Weight	Dept	Desc	Department	Description
Stability when equipment moves	10%	2	5	Every transfer changes the device's group, so its open schedule must be reassigned.	A transfer does not change what the device is, so its schedule is unaffected.
Traceability for engineers	10%	5	4	"ICU is done in January and July" is easy to say and to check.	Clear, but there are more rules to read.
Setup and administration effort	10%	4	2	Roughly 20 to 40 department rules per workshop.	Roughly 100 to 300 description rules, unless category templates are used (section 4).
Dependence on data quality	5%	4	3	Department assignment is usually clean.	Synonyms ("Patient Monitor" vs "Monitor, Patient") split groups.
Weighted score (out of 5)	100%	3.05	3.90	61%	78%

Sensitivity. Holding the other weights in proportion, department only wins once ward access carries more than about a third (34%) of the total. That is realistic for one case: a small hospital with one engineer doing rounds, where the number of trips matters more than batching. That is exactly the case the per-workshop switch is for.

When to choose which

Choose description when	Choose department when
Calibration, where one analyser and procedure serve a batch	Small workshop (under about 500 devices), one engineer doing rounds
Large workshop with many device types and several technicians	Wards with strict access windows (theatre, ICU, isolation)
Devices are often transferred between wards	Devices rarely move and departments are small
Intervals vary widely (3, 6, 12 months)	Most devices in a ward share one interval



3. Rating the current system

The existing code, rated against the fifteen business rules. The logic used (department or description) makes no difference here: both fail for the same reasons.

#	Business rule	Today	Why
1	One equipment, many completed schedules	Partial	Daily cleanup jobs delete completed history of inactive equipment.
2	Completed schedules are immutable	Fail	The PPM edit view can change a completed row's month and status; the validators delete them.
3	Only one pending schedule per equipment	Fail	No constraint. Only unique (equipment, month) exists.
4	Auto-scheduling never creates duplicates	Fail	Six writers each deduplicate only on their own target month.
5	Scheduling is deterministic	Fail	First-fit packing depends on the run date and row order; alignment snaps to the most common month.
6	Assignment through explicit groups	Partial	Department and description groups exist, but no one decides their months.
7	Groups have designated months	Fail	There is no month configuration at all.
8	Intervals are respected	Partial	Month + period is used, but the PPM push adds 30 days and the calibration clamp rebases on today.
9	Next month uses interval AND group months	Fail	Group months do not exist.
10	Logic change triggers controlled review	Partial	A warning is written, then everything is repacked, with no review step.
11	Completed never moved on logic change	Pass	Reorganisation excludes completed and locked rows.
12	Unscheduled equipment shown with a reason	Fail	"Unscheduled" means "never had any schedule", so broken chains are invisible.
13	Understandable at 5,000+ devices	Fail	No month view, no reason, no distribution.
14	Auto-scheduling is idempotent	Partial	A rerun skips equipment with any schedule, but the concurrent writers are not coordinated.
15	Every generated schedule is explainable	Fail	Only the code path that created a row is stored, and alignment overwrites it.
Score (Pass 1, Partial 0.5, Fail 0)		3.5 / 15	23%

The five defects behind most of the symptoms

- **PPM groups never advance.** `ppms/signals.py:120` marks a group as processed *before* checking whether it is complete. The first completion returns early, and every later completion is skipped. Only single-member groups ever get a next schedule.
- **One device blocks its group.** A group only advances when every member is completed. One broken or missing device stalls the whole group.
- **Broken chains are invisible.** "Unscheduled" means "never had a schedule", so equipment with only completed rows is never picked up or reported.
- **Months come from the run date.** First-fit packing with a limit of 100 groups per month puts each day's newly found equipment into whatever "next month" is.
- **History is deleted.** Two daily beat jobs delete every schedule, completed included, for inactive equipment.



4. The best version of each logic

Refinement to the deep dive: keep the interval and the months separate.

An interval describes a device type, so it is set **per description** (with a workshop default). The months are set **per group** of whichever logic the workshop chose. The engine then checks that every interval in a group fits that group's months. This one model works for both logics:

Group months	Device interval	Result
ICU: January, July	6 months	Visited January and July every year
ICU: January, July	12 months	Half the devices in January, half in July (balanced, deterministic), each once a year
ICU: January, July	3 months	Unschedulable, with a reason: "3-month interval needs 4 months; ICU has 2"
Patient Monitor: February, August	6 months	February and August every year

Making description logic practical

- **Category templates.** Set months once per category (for example Monitoring: February, August); every description inherits them unless overridden. That brings about 200 rules down to 15 or so, plus a handful of exceptions.
- **Monthly work lists sorted by department.** Grouping decides the month; inside the month, the job list is ordered ward by ward, which recovers most of department logic's routing advantage.
- **Description merge tool.** Merge synonyms before activating a plan, so one device type is never split across two rules.

Making department logic practical

- **Enough months for the shortest interval.** The rule editor warns when a department contains a device whose interval does not fit its months.
- **Transfers are handled explicitly.** A transfer reassigns the device's open schedule to the new department's next valid month, records why, and never touches completed history.
- **Large wards are split.** A department with more devices than a month can handle gets two months, and its 12-month devices are balanced between them.

For both logics

- Each device advances on its **own** completion; the group stays together because the months are fixed, not because devices wait for each other.
- Overdue is **shown**, not moved. The due month is permanent, so the cycle never drifts.
- A database constraint allows **one open schedule per equipment**.
- Every generated schedule stores **why**: rule, plan version, months, interval, previous schedule, chosen month.



5. Improvement plan

Each phase is useful on its own and can ship alone. Effort figures are rough estimates for one developer and include tests.

Phase	Goal	Effort	Risk	Work	Outcome
0	Stop the damage	~2 days	Low	Disable the history-deleting cleanup jobs. Fix the PPM processed-group bug. Stop the +30-day push. Define "unscheduled" as "no open schedule". Guard completed rows in save().	Rules 1, 2 pass; stalled PPM groups advance again
1	Audit and constraint	~3 days	Medium	Read-only audit report. Deactivate (never delete) duplicate open schedules after review. Add the one-open-schedule constraint locally and on HQ.	Rules 3, 4 pass
2	Plans and engine	~5 days	Medium	Per-workshop, versioned plans with department or description logic. Per-description intervals. One deterministic engine and one nightly sweep; retire the old writers.	Rules 5-9, 14, 15 pass
3	Administrator screens	~5 days	Low	Plan editor with preview and apply. Month view. Unscheduled report with reasons. Schedule history and reason on the equipment page. Distribution analytics.	Rules 10, 12, 13 pass
4	Rollout	~2 days + 1 cycle	Medium	"Adopt current layout" per workshop, review, activate. Watch the monthly distribution for one cycle.	15 / 15

Expected score after each phase

	Today	After 0	After 1	After 2	After 3	After 4
Business rules met (of 15)	3.5	5.5	7.5	13	15	15
Share	23%	37%	50%	87%	100%	100%

Phase 0 changes no data model and no screens, so it is safe to ship on its own this week. It is also the phase that most directly explains "equipment remains unscheduled".

6. Still open

Question	Recommendation
Per-equipment interval override (one special ventilator on 3 months)?	Not at first. Add it later if a real case appears; it adds a synced field to Equipment.
Late completion: next slot after the due date, or never exceed the interval?	Calibration: never exceed (certificate validity). PPM: next slot after the due date.
Remove group-completion gating?	Yes. It is the main cause of unscheduled equipment.
Who runs auto-scheduling across the fleet?	Every desktop, made safe by deterministic row IDs and the database constraint.
Live data check	Run the audit once the dev database (port 2215) is up. It is in a restart loop and was not reachable for this review.